1 Initial Method and Experiment Details

We consider a simplified version of our multi-agent framework to establish a strong baseline. The full system consists of three LLM agents:

1. Example Generator
2. Relation Inference Agent
3. Feedback Agent

In the current setting, we evolve only the prompt of the Relation Inference Agent (Agent 2) using feedback from the Feedback Agent (Agent 3). The Example Generator (Agent 1) is not used. No additional in-context examples are introduced. All experiments are therefore conducted under a **1-shot baseline** configuration.

We update the inference prompt using structured feedback generated by a separate feedback prompt with three variants:

- **Corrects**,
- **Mistakes**, and
- **Corrects + Mistakes**.

Let $D_{feedback}$ denote the feedback (training) set. Let $D_{validation}$ denote the validation set. Let D_{test} denote the test set. We initialize the system as S_0 with a set of prompts $P = \{P_{inference}, P_{feedback}, P_{mutation}\}$. Only the inference prompt $P_{inference}$ is subject to evolution. The initial population is defined as $Q = \{S_0\}$.

At each iteration, we perform the following steps for `MAX_ITERATION` $= 20$.

1. Sample a system S_x from the population Q based on its validation performance (see Section 1.1).
2. Sample f instances from $D_{feedback}$, with $f = 3$ by default.
3. Run relation inference on the sampled instances using the inference prompt $P_{inference}$ of S_x.
4. Collect structured feedback on the model outputs using a fixed evaluation prompt $P_{evaluation}$.
5. Use $P_{mutation}$ to update $P_{inference}$ based on the structured feedback, yielding a new system S'_x.
6. Evaluate S'_x on the validation set $D_{validation}$.
7. Add S'_x to the population Q.

After all iterations, we select the system in Q with the highest validation performance. The selected system is then evaluated on the test set D_{test}.

Models and Dataset: We conduct experiments using **Qwen3-4B** and **Gemma3-4B** as the underlying language models. All experiments are performed on the **FS-TACRED** dataset.

Data Format: The FS-TACRED validation and test sets follow an episodic few-shot format with a support set and a query. The validation episodes are constructed from the original validation split. The feedback set (using the training split of FS-TACRED) is not episodic; we sample two instances, one serving as a 1-shot support example that defines the relation and the other serving as the query.

1.1 System Sampling

Let the population be $Q = \{S_1, S_2, \ldots, S_{|Q|}\}$, where each system S_i has a validation score v_i. To balance exploration and exploitation, we sample a system from Q according to a softmax distribution over validation scores:

$$p(S_i) = \frac{\exp(v_i/\tau)}{\sum_{j=1}^{|Q|} \exp(v_j/\tau)},$$

where $\tau > 0$ is a temperature parameter controlling exploration; we initialize $\tau = 1$.